

Meeting Summaries of April 2

Date: April 2

Time: 18:00 - 18:30 (PDT)

Attendees: Prof. Jan, Dr. Javed, Dr. Yue and Feng

Topic: Progress report on **Chapter 2** and **Chapter 3**; discussion of **RDA methodology**, permutation tests, p-values and interpretation of key figures; planning of additional in-person meetings and April work schedule.

Questions, Feedbacks and Suggestions

- Dr. Javed and Prof. Jan: Before using the updated **RDA** results in the thesis, Feng should explain the full procedure in plain words. The explanation should state the **hypothesis**, what was permuted in the permutation test, how the pseudo- F statistic and p-value were obtained, and how these quantities were used to judge the candidate cut-off values for reference sites.

Key point for the next meeting

Before writing more interpretation, prepare a short explanation for **each important figure**: what data entered the analysis, what statistical question was tested, what appears on each axis, and how the reported statistic supports the conclusion.

- Prof. Jan: For each important figure, Feng should write a self-contained caption or short explanation that states how the figure was created, what is shown on the x-axis and y-axis, and what each line, point or panel represents. The figures should be understandable even without oral explanation.
- Prof. Jan and Dr. Javed: The statistical significance in the RDA results should be interpreted carefully. Feng should distinguish between changes caused by the percentage of selected reference sites and changes caused by the absolute sample size N , because p-values can become smaller when sample size increases even if the underlying effect changes only slightly.
- Prof. Jan: Before extending the **HZD**-based contamination assessment, Feng should review the referenced paper and clarify the meaning of **TEL** and **ERL**, etc, in the HZD metric, and explain how these toxicity benchmarks relate to contamination assessment in this thesis.
- Dr. Javed: Starting on April 15, Feng will meet with Dr. Javed and Dr. Yue in person every two days, in addition to the regular committee meeting, to review progress and statistical analysis. Feng should book a room and coordinate meeting times that work for everyone.

- Dr. Yue: Progress on the genetic-data graph work has been slower than expected. Feng should set aside specific hours each week in April for focused work, communicate that schedule to Dr. Yue, and complete the current graph analysis before adding **g_loss** and **gn_rotation** to the model.
- Prof. Jan: The missing data port from Toronto is expected early next week. Once it arrives, Prof. Jan will process it and notify the team.

Summary of Discussion

- Feng opened the meeting with updates about the **SSC 2026 Conference**. His abstract was accepted for an oral presentation, the submitted summary was selected for student presentation awards, and most of the travel cost will be covered by grants.
- Feng then reported progress on **Chapter 2**. Over the past week, he summarized the paper on hazard metrics for benthic invertebrate toxicity, applied **HZD metrics** to the available chemicals with consensus benchmarks, and updated the **RDA** analysis used to identify candidate cut-off values for least polluted reference sites. The new RDA figures compared environmental predictors and contamination stressors using R^2 , pseudo- F , p-values and VIF.
- The main discussion centered on whether the updated RDA methodology and its interpretation were fully understood. Dr. Javed and Prof. Jan questioned Feng's understanding of the permutation test, the meaning of the p-values, and the statistical significance of the results. They emphasized that the figures should support a clear explanation of what was tested, why it was tested, and what conclusion can be defended.
- The meeting also revisited the contamination assessment results from **SumRel**, **MaxRel**, and the **HZD**-based approach. Feng noted that some rankings and patterns differed because of the scoring and standardization procedures. The committee asked him to understand these differences more clearly before treating the results as final, especially the meaning of **TEL** and **ERL** in the HZD framework.
- Feng also reported progress on **Chapter 3**. The clustering work now uses **Ward's clustering** to define taxa groups, with MRT treated as a candidate classifier, and several dense figures were separated into simpler plots to improve readability. The updated Detroit River analysis, which includes water velocity, produced better classification results than the larger corridor case.
- The final part of the meeting addressed scheduling and project management. Starting in mid-April, additional in-person meetings will be arranged for progress updates and statistical review. Feng will prepare a weekly April work schedule with specific hours for focused work, while the supervisors noted that summer travel will reduce flexibility later, so key methods and calculations should be clarified as early as possible.

Action Items

- Feng will write clear explanations and captions for the most important figures, describing how each figure was created, what is shown on the x-axis and y-axis, and what every major component represents. These notes will be shared with the committee for feedback before the next meeting.

- Feng will supplement and clarify the hypothesis and detailed steps of the permutation test and p-value calculation in the RDA analysis, and present this explanation in the next meeting.
- Feng will update the analyses to include the absolute sample size N in addition to the percentage in a locally total observation of reference sites, and report whether the observed changes are driven by proportion, sample size, or both.
- Feng will review the referenced paper to understand the meaning of **TEL** and **ERL** in the HZD metric and explain how they should be interpreted in relation to the contamination assessment.
- Feng will check rooms for the additional in-person meetings starting on April 15, and coordinate with Dr. Javed and Dr. Yue so that the schedule works for everyone.

Follow-Up

Temporary meeting time: April 9, 18:00 - 19:00 (PDT)

Supportive materials

- A meeting recording has been restored on local drive.